

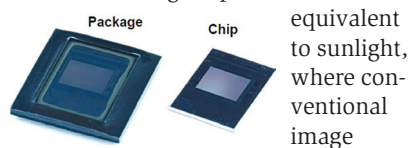
OEMs and Tier 1 suppliers can now control multiple power and energy functions simultaneously in real time. Early figures show up to 5% energy efficiency improvement, 25% motor downsizing for the same power, 35% less cooling need, and up to 30 times passive component downsizing compared to current EVs.

Silicon Mobility

<https://www.silicon-mobility.com>

3D TOF sensor with integrated distance calculation circuit

Nuvoton Technology Corporation, Japan has launched a 6.35mm VGA resolution 3D time-of-flight (TOF) sensor. The sensor measures distances accurately even in environments with ambient light up to 100,000 lux,



equivalent to sunlight, where conventional image sensors overexpose. Equipped with a built-in distance calculation circuit and a signal correction ISP, the sensor outputs distance (3D) images at up to 120 frames per second (fps) without delay, along with IR (2D) images for enhanced recognition accuracy. The 3D TOF sensor provides precise recognition of the position and shape of people and objects in any environment. This improves safety, security, and efficiency in automotive, consumer applications like smart door locks, and industrial uses like cargo measurement in logistics.

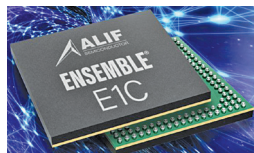
Nuvoton Technology Corporation

<https://www.silicon-mobility.com>

MCU series with AI and low-power features

Alif Semiconductor has introduced the Ensemble E1C, an MCU series combining digital and analogue capabilities with a low-power on-chip neural processing unit (NPU) in a compact form factor. The MCU features a 160MHz ARM Cortex-M55 CPU core with Helium vector processing extension

and an ARM Ethos-U55 NPU delivering up to 46 GOPs. The MCU provides an entry-level option for OEMs seeking high AI performance and low



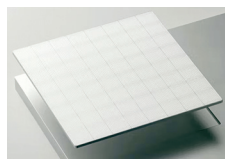
power consumption in an ARM environment. The E1C enables both AI/ML and application control functions at low power levels, suitable for products like wearable devices with strict power and space constraints.

Alif Semiconductor

<https://alifsemi.com>

100x faster CPUs with new PPU addition

Finnish startup Flow Computing has launched a coprocessor that can be added to a CPU to deliver 'up to 100-fold' acceleration for highly-parallelisable workloads—without modifying existing software or the high power demands of traditional GPUs



and accelerators. The company claims that the performance gains achieved through the integration of its PPU technology into a CPU design are "fully backwards compatible with every existing software application for that CPU architecture," requiring only 'recompilation for the PPU without any software changes.' The company says its design, tailored for highly-parallel workloads, avoids common pitfalls of rival accelerators such as cache coherence issues, inefficient flow synchronisation, and provides solutions for concurrent memory access.

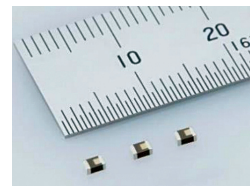
Flow Computing

<https://flow-computing.com>

World's first multilayer metal power inductor

Taiyo Yuden Co., Ltd has a multilayer metal power inductor MCOIL 'LAC-NF2012KKTR24MAB', which meets the AEC-Q200 reliability qualification for passive automotive components.

The company claims to achieve an upper operating temperature limit of 165°C for a multilayer metal power inductor, a world first. It can be used



as choke coils in DC-DC converters for automotive engine ECUs, control systems

like BMS, safety systems like ABS, body-related systems like ADAS, and information systems like instrument clusters. The inductor contributes to the miniaturisation and performance enhancement of power supply circuits, enabling high-density mounting in high-temperature environments such as automobiles.

Taiyo Yuden Co., Ltd

<https://www.yuden.co.jp/ap>

Open standard module for embedded computing

ADLINK Technology has introduced the open standard module (OSM), a compact and powerful form factor transforming embedded computing. OSM modules set a new benchmark in



compact, open standard, and solderable BGA mini modules, supporting

both ARM and x86 designs while being significantly smaller than any available computer-on-modules. The OSM redefines size efficiency, with the largest module measuring only 45mm x 45mm, making it 28% smaller than Qseven and 51% smaller than SMARC. The BGA design allows for more interfaces on a smaller footprint, crucial for miniaturisation and meeting complex requirements. With a power envelope typically under 15W and the durability to withstand extreme vibration, OSM is suitable for applications requiring resilience in rugged environments.

ADLINK Technology

<https://www.adlinktech.com>